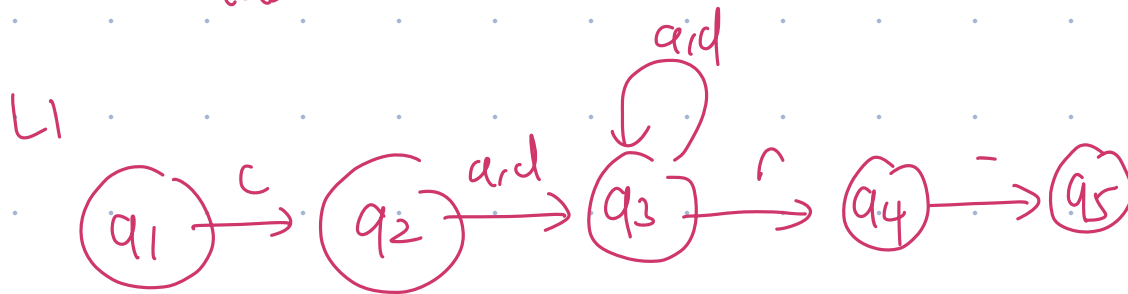
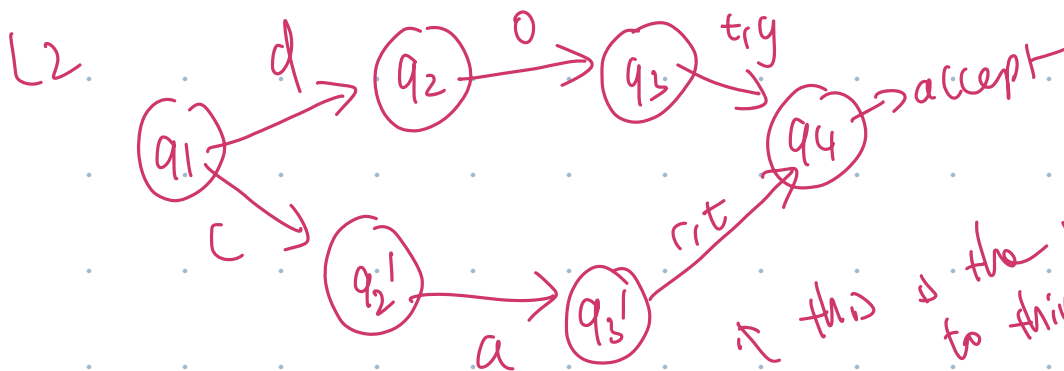


The Automata I designed has the bug that "cr" would also be accepted.



the way I thought about it is not correct



this is the way to think about it.

A language that can be decided by a DFA is called regular.

Most text editors (find & replace: regular expressions) function

More language related to language L .

① complement $\overline{L}$

② Kleene closure: concatenations of 0 or more strings in L .

Is the complement of a regular language also regular?

↳ YES: change accept $\Rightarrow$ reject
reject $\Rightarrow$ accept

Does closure a regular language?

e^*

↳ YES.

More languages.

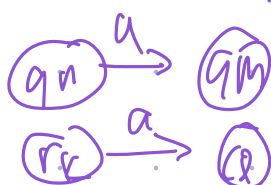
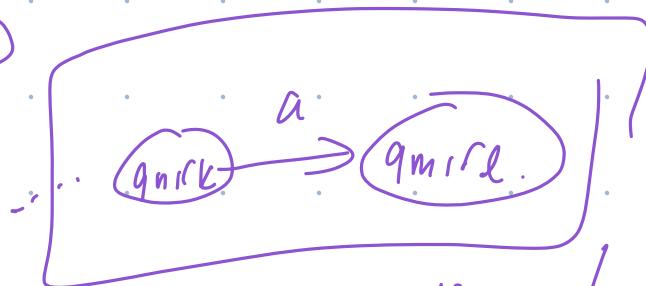
① Union

② Intersection

③ Concatenation: all string formed by appending a string in L_1 to one in L_2 .

~~Union~~

How to build a FSM that simulates both languages (to check if string $\in$ union?)



tracking simultaneously any state in the 1st & 2nd machine.

NFA $\rightarrow$ wasn't paying attention
to a
 $\downarrow$ DFA

Order of operations are like arithmetic:

- $\hookrightarrow$ Concatenation: times
- $\hookrightarrow$ Union: plus
- $\hookrightarrow$ Star: exponentiation.

Theorem: For any regular expression e ,
 $L(e)$ is a regular language.

Proof: Single letters can be recognized by a very simple DFA. Unions and Kleene closures of regular languages are regular.

A string of a's and b's containing an even number of a's.

$b^*(ab^*ab^*)^*$

nice construction!

I feel like you still need

b^* at the end.

$(aa)(aa)b^4$

Palindromes / Fibonacci number length.

You can actually use $re(g|u)$.

↖ then

notation to do search in
VScode.

or $99[0-9]$

↳ 990, 991, ... 999.

$([A-Z])([a-z]^*)$

↑
\$1

first group

↑
\$2

2nd group